CodeEmy™
TECHNOLOGIES
Ignite the Flame of Education

Clinical Data Management & EDC

Master the Clinical Data Management.

Duration: 3 Months | Mode: Online | Level: Beginner to Advanced
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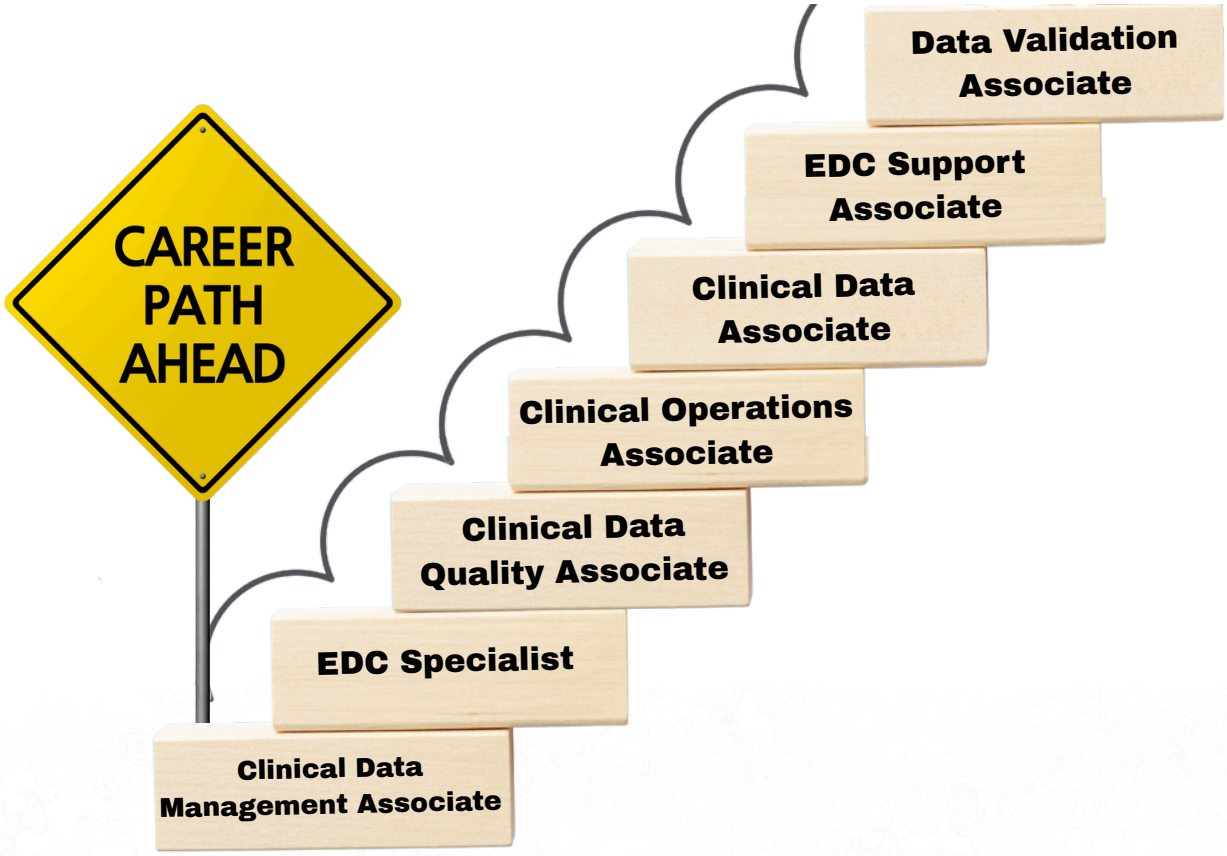
Scope of
Clinical Data Management & EDC Industry

- Clinical Trial Data Management
- Electronic Data Capture (EDC)
- Clinical Research & CRO Operations
- Clinical Database Management
- Clinical Data Validation
- Data Cleaning & Query Management
- Clinical Data Quality & Reconciliation
- Clinical Trial Data Standards
- Clinical Database Lock
- Healthcare & Pharmaceutical Data Operations

What Students Will Learn

Clinical Research → CRF → EDC → Data Entry → Validation → Query Management → Data Cleaning → Data Reconciliation → CDISC → Database Lock

Here is all the Career Fields you can excel at



CAREERS

CodeEmy TECHNOLOGIES

Overview:

Curriculum

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Week 1

- Introduction to Clinical Research, Drug Development Process
- Clinical Trial Phases, Sponsor, CRO & Clinical Sites
- Introduction to Clinical Data Management
- Role of Clinical Data Management in Clinical Trials
- Clinical Trial Data Lifecycle, Introduction to EDC

Project: Map the Clinical Trial Data Lifecycle

Week 2

- Clinical Trial Protocol & Case Report Forms (CRF)
- Source Data & Source Documents & Patient Demographics
- Medical History & Concomitant Medications & Adverse Events & Laboratory Data
- Vital Signs & Clinical Assessments & Data Collection Principles

Project: Design a Sample Clinical CRF

Week 3

- Introduction to Electronic Data Capture
- EDC Architecture & Workflow & User Roles & Access
- Study Setup & Site & Subject Management
- Forms & Visit Structure & Data Entry & Data Modification
- Audit Trail Concepts & EDC Workflow Simulation

Project: Build a Simulated EDC Study Structure

Week 4

- CRF Design Principles & Paper CRF vs Electronic CRF
- Data Fields & Variables, Field Types & Mandatory & Optional Fields
- Edit Checks, Range Checks & Data Validation Rules
- Database Structure & Visit & Form Mapping

Project: Create a Clinical Database / CRF Design

Week 5

- Clinical Data Entry & Double Data Entry Concepts
- Data Verification, Data Validation & Missing Data
- Inconsistent Data, Out-of-Range Data & Data Review
- Error Identification & Data Correction
- Project: Clinical Data Entry & Validation Case Study

Week 6

- Introduction to Clinical Data Queries
- Query Identification, Creation, Assignment & Resolution
- Query Response, Closure & Common Data Issues
- Query Tracking & Query Metrics

Project: Create & Resolve Clinical Data Queries

Week 7

- Clinical Data Cleaning & Data Review Process
- Missing & Inconsistent Data & Duplicate Records
- Cross-Form Validation & External Data Reconciliation
- Laboratory Data & Safety Data & Medication Data
- Data Cleaning Documentation

Project: Clinical Data Cleaning Exercise

Week 8

- Introduction to CDISC & Clinical Data Standards Overview
- CDASH Fundamentals & SDTM Fundamentals
- Variables & Domains & Data Mapping Concepts
- Controlled Terminology & Standardized Clinical Data
- Introduction to ADaM

Project: Map Sample Clinical Data to CDISC Structure

Week 9

- Introduction to Medical Coding in Clinical Trials, MedDRA Overview & Adverse Event Coding
- Medical History Coding & Indication Coding
- WHODrug Overview & Concomitant Medication Coding & Coding Review
- Coding Queries & Coding Reconciliation

Project: Clinical Event & Medication Coding Case Study

Week 10

- Clinical Data Review & Data Review Listings
- Data Validation & Safety Data Reconciliation
- Laboratory Data Reconciliation & External Data Transfer
- Data Quality Checks & Protocol Deviations & Data Impact
- Data Integrity & Quality Control
- Project: Clinical Data Quality & Reconciliation Report

Week 11

- Database Freeze, Database Lock & Pre-Lock Activities
- Final Data Cleaning & Outstanding Query Review
- Data Reconciliation& Final Data Review
- Database Lock Checklist & Data Transfer & Archiving
- Clinical Data Management Documentation

Project: Prepare a Database Lock Readiness Checklist

Week 12

- End-to-End Clinical Data Management Workflow
- EDC Data Entry Simulation & Data Validation
- Query Management & Data Cleaning
- Medical Coding & Reconciliation
- CDISC Mapping & Database Lock
- Resume Preparation & Interview Preparation

Project: Complete Clinical Data Management Capstone

Hands - On Projects

Mini Projects (Beginner to Intermediate Level)

1. Clinical Trial Data Lifecycle Mapping
Skills Used: Clinical research, clinical data management, data lifecycle
Goal: Map the complete flow of clinical trial data from Patient → CRF → EDC → Validation → Cleaning → Database Lock.

2. Clinical CRF & EDC Form Design
Skills Used: CRF design, data fields, variables, EDC concepts
Goal: Design sample electronic CRFs for patient demographics, medical history, laboratory data and adverse events.

3. Clinical Data Query Management
Skills Used: Data validation, query creation, data review
Goal: Identify data inconsistencies and create, track and resolve simulated clinical data queries.

Major Projects (Advanced Level)

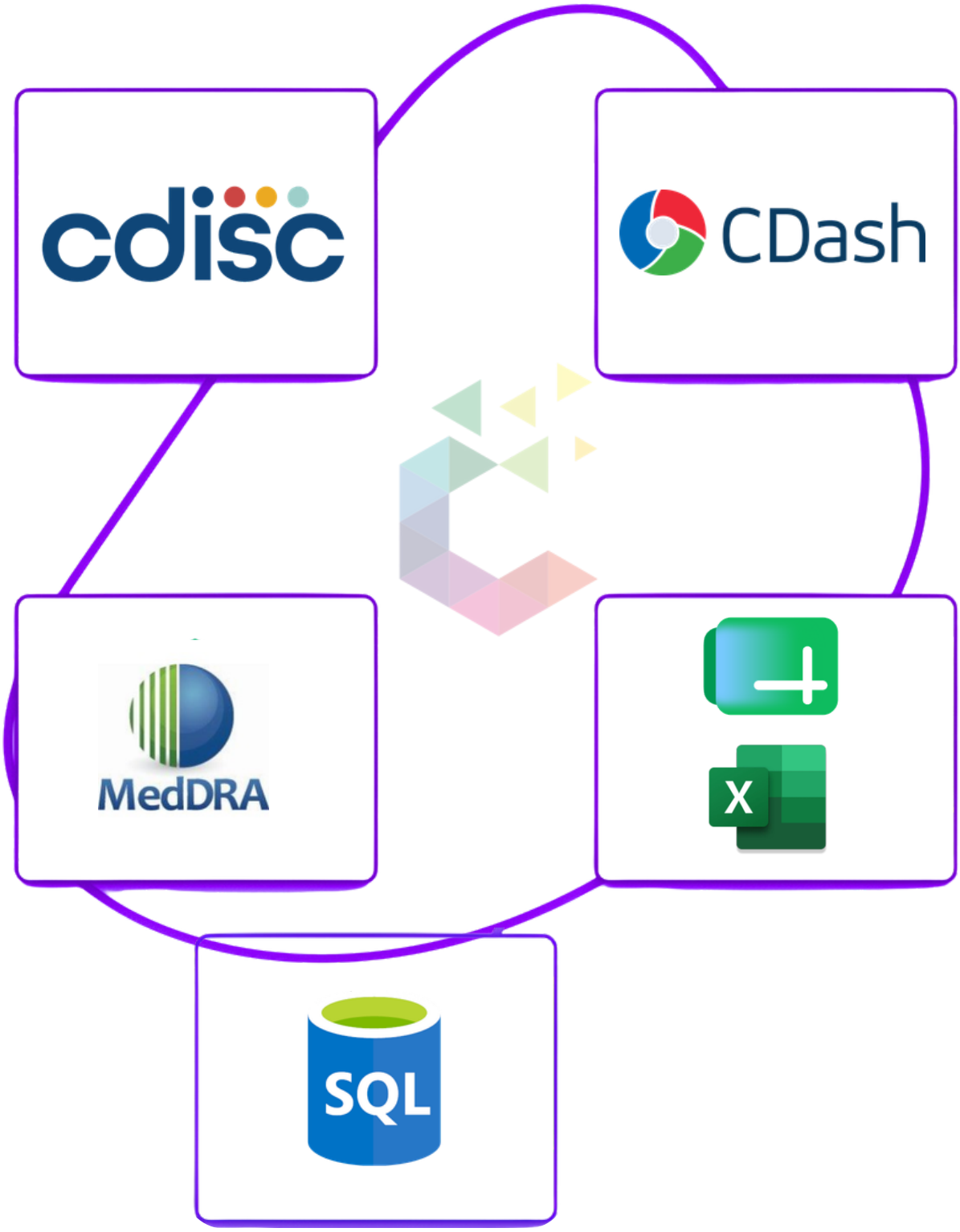
1. Complete Clinical Trial EDC Study Setup
Skills Used: EDC, CRF design, database structure, data entry, validation
Goal: Build a simulated clinical study with study structure → visits → CRFs → subjects → data entry → validation.

2. Clinical Data Cleaning & Reconciliation Project
Skills Used: Excel, data cleaning, query management, reconciliation, data quality
Goal: Analyse a clinical dataset, identify discrepancies, generate queries, perform data cleaning and prepare a final data-quality report.

3. End-to-End Clinical Data Management Capstone
Skills Used: EDC, CRF, data entry, validation, queries, MedDRA, CDISC, reconciliation, database lock
Goal: Execute a complete simulated clinical data management workflow from CRF Design → EDC → Data Entry → Validation → Queries → Cleaning → Coding → Reconciliation → CDISC Mapping → Database Lock.

Tools

You will learn



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your learning journey

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Certificate of Internship Completion from CODEEMY

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Contact Us

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